

Ayantanu Laha

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PROFESSIONAL SUMMARY

Software Engineer with **1+ year** of experience specializing in **Data Engineering and AI solutions**. Gold Medalist in M.Sc Computer Science. Proven track record building scalable Firebase/BigQuery pipelines serving 25,000+ users.

PROFESSIONAL EXPERIENCE

Software Engineer

June 2025 – Present

SenSight Technologies Pvt. Ltd., Bangalore, India

- Architected and deployed a **Firebase-based notification infrastructure** serving **25,000+ active users** across Android (FCM) and iOS (APNs), implementing silent push notifications and scheduled delivery with a **99.5% success rate**
- Designed and developed an end-to-end **in-app Rewards Platform**, building a **Python-based reward computation engine** for scoring, incentive calculation, and eligibility logic, alongside scalable **Spring Boot REST APIs** for real-time rewards, balances, and redemption workflows
- Implemented **BigQuery data pipelines** processing **100k+ records daily** and built **15+ interactive enterprise dashboards** in **Amazon QuickSight** and **Looker Studio**, integrating diverse data sources with **role-based access control (RBAC)** to improve operational visibility and decision-making for multi-country clients

Software Engineer Intern

January 2025 – June 2025

SenSight Technologies Pvt. Ltd., Bangalore, India

- Developed an **AI-powered driver behavior analysis system** using **OpenAI**, generating personalized safety reports for **10,000+ drivers** with **85% positive feedback** on actionable insights
- Created **custom Amazon QuickSight dashboards** with advanced KPI visualizations, reducing manual reporting time by 40% for operations teams

PROJECTS

RAG-Enhanced Cross-Lingual Retrieval: Bridging Bilingual Query Gaps

- Tools & Technologies: Pandas, NumPy, FAISS DB, Hierarchical Navigable Small World (HNSW) Algorithm, Hugging Face, LLaMA 3, Streamlit.
- Developed a cross-lingual Retrieval-Augmented Generation (RAG) system using FAISS as the vector database.
- Designed a cross-lingual retrieval system, improving search relevance for Bengali & English queries.
- Created relational mappings between Bengali and English content for improved cross-lingual retrieval.
- **Link to Project:** GitHub

TECHNICAL SKILLS

- **Languages:** Python (**Proficient**), C++, Java
- **Cloud & Data:** BigQuery, Firebase, GCP, Amazon QuickSight, MySQL
- **ML & AI:** PyTorch, Hugging Face, LLaMA, OpenAI API, FAISS, RAG Systems
- **Data & Visualization:** Pandas, NumPy, GeoPandas, Shapely, Looker Studio
- **Tools:** Streamlit, HERE Maps, Folium

EDUCATION

Ramakrishna Mission Vivekananda Educational and Research Institute

Howrah, India

M.Sc in Computer Science

2023 – 2025

- CGPA: **8.11/10.0**
- **Gold Medalist** — Ranked **1st** in Computer Science
- Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Database Management Systems, Data Structures & Algorithms

Ramakrishna Mission Residential College, Narendrapur

Kolkata, India

B.Sc in Computer Science

2020 – 2023

- CGPA: **9.61/10.0**
- Graduated with **First Class** with Distinction
- Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networking, Theory of Computation, Computer Organization & Architecture